Graphing Inequalities Error Analysis TEACHER KEY

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Solid or dashed?

ORIGINAL PROBLEM

Graph $y > 2x - 1$

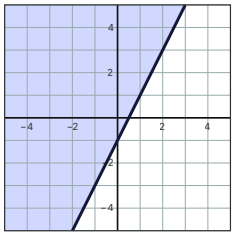
INCORRECT WORKED SOLUTION

$y > 2x - 1$

Slope 2, y-intercept −1.

Draw a SOLID line.

Shade above the line.



A. EXPLAIN THE MISTAKE (1 POINT)

The symbol > does not include equal to, so points on the line are NOT solutions. The boundary must be a DASHED line.

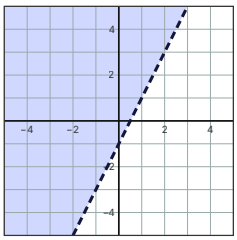
B. SHOW CORRECT WORK (1 POINT)

$y > 2x - 1$

Slope 2, y-intercept −1.

Draw a DASHED line.

Shade above the line.



2. Solid or dashed?

ORIGINAL PROBLEM

Graph $y \leq -x + 3$

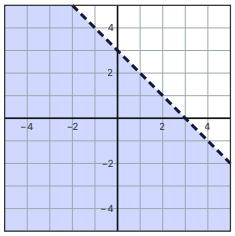
INCORRECT WORKED SOLUTION

$y \leq -x + 3$

Slope −1, y-intercept 3.

Draw a DASHED line.

Shade below the line.



A. EXPLAIN THE MISTAKE (1 POINT)

The symbol ≤ includes equal to, so points on the line ARE solutions. The boundary must be a SOLID line.

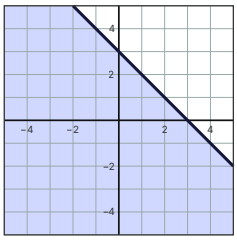
B. SHOW CORRECT WORK (1 POINT)

$y \leq -x + 3$

Slope −1, y-intercept 3.

Draw a SOLID line.

Shade below the line.



Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Which side to shade

ORIGINAL PROBLEM

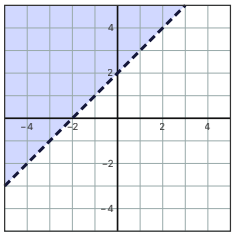
Graph $y < x + 2$

INCORRECT WORKED SOLUTION

$y < x + 2$

Dashed line through (0, 2) with slope 1.

Shade above the line.



A. EXPLAIN THE MISTAKE (1 POINT)

$y <$ means y-values LESS than the line, so shade BELOW. Check with (0, 0): $0 < 2$ is true, and (0, 0) is below the line.

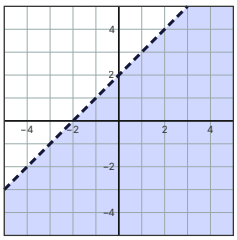
B. SHOW CORRECT WORK (1 POINT)

$y < x + 2$

Dashed line through (0, 2) with slope 1.

Test (0, 0): $0 < 2$ is true.

Shade below the line (the side with (0, 0)).



4. Vertical boundary

ORIGINAL PROBLEM

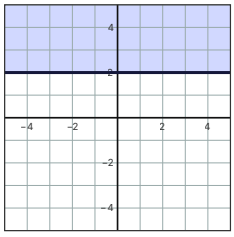
Graph $x \geq 2$

INCORRECT WORKED SOLUTION

$x \geq 2$

Draw a solid horizontal line through $y = 2$.

Shade above the line.



A. EXPLAIN THE MISTAKE (1 POINT)

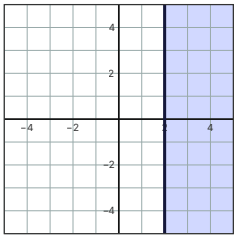
$x \geq 2$ is about x-values. The boundary $x = 2$ is a VERTICAL line, and the solutions are to the RIGHT of it.

B. SHOW CORRECT WORK (1 POINT)

$x \geq 2$

Draw a solid VERTICAL line through $x = 2$.

Shade to the right of the line.



Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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5. Plotting the slope

ORIGINAL PROBLEM

Graph $y \geq \frac{1}{2}x - 3$

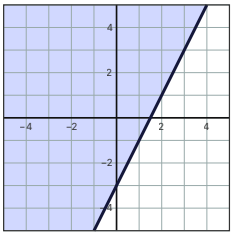
INCORRECT WORKED SOLUTION

$y \geq \frac{1}{2}x - 3$

Start at $(0, -3)$.

Slope $\frac{1}{2}$: go up 2, right 1 to plot $(1, -1)$.

Solid line, shade above.



A. EXPLAIN THE MISTAKE (1 POINT)

Slope $\frac{1}{2}$ means RISE 1, RUN 2: up 1 and right 2, to the point $(2, -2)$. The wrong graph went up 2 and right 1, which is slope 2.

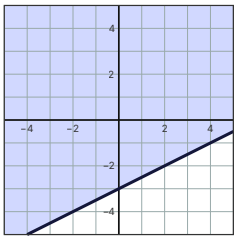
B. SHOW CORRECT WORK (1 POINT)

$y \geq \frac{1}{2}x - 3$

Start at $(0, -3)$.

Slope $\frac{1}{2}$: go up 1, right 2 to plot $(2, -2)$.

Solid line, shade above.



6. Reading the y-intercept

ORIGINAL PROBLEM

Graph $y < -2x + 4$

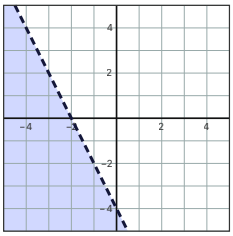
INCORRECT WORKED SOLUTION

$y < -2x + 4$

Start at $(0, -4)$.

Slope -2: down 2, right 1.

Dashed line, shade below.



A. EXPLAIN THE MISTAKE (1 POINT)

The y-intercept is +4, so the line starts at $(0, 4)$, not $(0, -4)$. Then go down 2 and right 1 to $(1, 2)$.

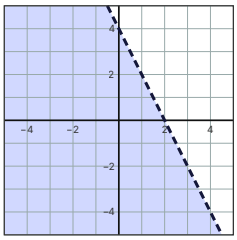
B. SHOW CORRECT WORK (1 POINT)

$y < -2x + 4$

Start at $(0, 4)$.

Slope -2: down 2, right 1 to $(1, 2)$.

Dashed line, shade below.



Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Graphing Inequalities Error Analysis TEACHER KEY

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7. Rewriting first

ORIGINAL PROBLEM

Graph $2x + y \leq 4$

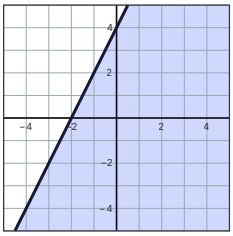
INCORRECT WORKED SOLUTION

$2x + y \leq 4$

$y \leq 4 + 2x$

$y \leq 2x + 4$

Solid line through (0, 4), slope 2; shade below.



A. EXPLAIN THE MISTAKE (1 POINT)

To get y alone, SUBTRACT 2x from both sides: $y \leq -2x + 4$. The slope is -2, not 2.

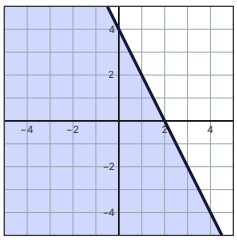
B. SHOW CORRECT WORK (1 POINT)

$2x + y \leq 4$

$y \leq -2x + 4$

Solid line through (0, 4), slope -2.

Shade below.



8. Using a test point

ORIGINAL PROBLEM

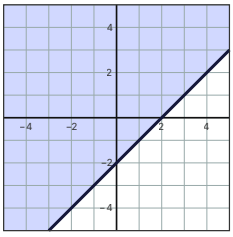
Graph $y \leq x - 2$

INCORRECT WORKED SOLUTION

$y \leq x - 2$

Test (0, 0): $0 \leq -2$ is false.

Shade the side that contains (0, 0).



A. EXPLAIN THE MISTAKE (1 POINT)

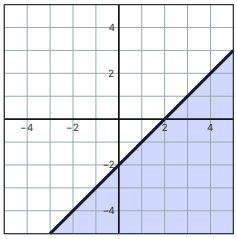
If the test point makes the inequality FALSE, that side is NOT in the solution. Shade the OTHER side: below the line.

B. SHOW CORRECT WORK (1 POINT)

$y \leq x - 2$

Test (0, 0): $0 \leq -2$ is false.

Shade the side WITHOUT (0, 0): below the line.



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9. Horizontal boundary

ORIGINAL PROBLEM

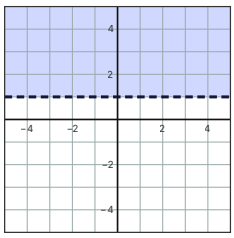
Graph $y > -1$

INCORRECT WORKED SOLUTION

$y > -1$

Draw a dashed horizontal line through 1 on the y-axis.

Shade above the line.



A. EXPLAIN THE MISTAKE (1 POINT)

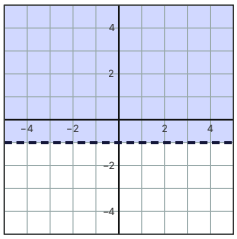
The boundary is $y = -1$, which is BELOW the x-axis. The wrong graph put the line at $y = 1$.

B. SHOW CORRECT WORK (1 POINT)

$y > -1$

Draw a dashed horizontal line through -1 on the y-axis.

Shade above the line.



10. Bad test point

ORIGINAL PROBLEM

Graph $y > x$

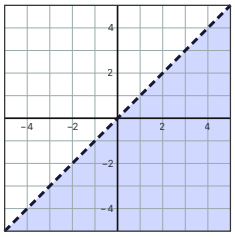
INCORRECT WORKED SOLUTION

$y > x$

Dashed line through the origin with slope 1.

Test (1, 1): $1 > 1$ is false.

Shade below the line.



A. EXPLAIN THE MISTAKE (1 POINT)

The point (1, 1) is ON the boundary line, so it cannot be a test point. Choose a point off the line, such as (0, 2): $2 > 0$ is true, so shade ABOVE.

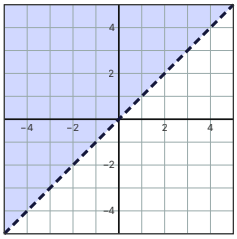
B. SHOW CORRECT WORK (1 POINT)

$y > x$

Dashed line through the origin with slope 1.

Test (0, 2): $2 > 0$ is true.

Shade above the line.



Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points